

Criterion: Database Design & Data Integrity

Architecture Decision Record

Status

Status: Accepted

Date: 2026-01-05

Context

The platform stores structured disease content (diseases, symptoms, risk factors, sources) with bilingual support (EN/RU) and many-to-many relationships. The schema must enforce integrity while staying easy to evolve during thesis development.

Decision

Use PostgreSQL with Prisma and a normalized schema: core entities (Disease, Symptom, RiskFactor, Source), junction tables for M:N relationships with metadata, and translation tables keyed by locale.

Alternatives Considered

Alternative	Pros	Cons	Why Not Chosen
Single denormalized diseases table	Very fast to build	Poor integrity; harder filtering; harder to scale	Doesn't scale for symptom/risk-factor relationships
Document DB (MongoDB)	Flexible schema	Harder relational queries/integrity	Data is inherently relational and benefits from SQL
Raw SQL access	Full control	Higher maintenance; harder type safety	Prisma provides migrations and type-safety

Consequences

Positive: - Strong referential integrity (FKs + cascades) - Expressive filtering/search across relationships - Locale support without duplicating core records

Negative: - More tables and joins than a single-table MVP

Neutral: - Translation tables can be extended to additional locales later

Implementation Details

Project Structure

```
backend/prisma/
+-- schema.prisma      # Canonical schema
+-- migrations/         # Prisma migration history
```

```
+- seed.ts # Seed script
```

Key Implementation Decisions

Decision	Rationale
Junction tables for M:N	Enables per-link metadata (priority/t...
Translation tables	Locale-aware fields without duplicati...
Enum types	Prevent invalid values and simplify A...
Uniques + indexes	Enforce data integrity and speed filt...

Code Examples

Locale enum + translation uniqueness:

```
enum Locale { en ru }

model DiseaseTranslation {
  diseaseId Int
  locale Locale
  @@unique([diseaseId, locale])
}
```

M:N with metadata:

```
model DiseaseSymptom {
  diseaseId Int
  symptomId Int
  priority SymptomPriority
  typicality Typicality
  @@unique([diseaseId, symptomId])
}
```

Diagrams

* The schema is also represented in the project's architecture/ERD diagrams under docs/assets/diagrams/.

Requirements Checklist

#	Requirement	Status	Evidence/Notes
1	Relational DB suitable for structured data	[OK]	docker-compose.yml uses postgres:16
2	Type-safe ORM layer with migrations	[OK]	backend/prisma/schema.prisma, Prisma ...
3	Supports bilingual (EN/RU) content	[OK]	Translation tables + Locale enum
4	Supports M:N disease?symptom?with met...	[OK]	DiseaseSymptom (priority/typicality)
5	Supports M:N disease?risk factor?with...	[OK]	DiseaseRiskFactor (RiskDirection)
6	Integrity constraints and indexes	[OK]	@@unique, @@index in schema

7	Sources stored for attribution	[OK]	Source model present
8	Seed process available	[OK]	backend/prisma/seed.ts and npm run seed

Known Limitations

Limitation	Impact	Potential Solution
Source is not linked to diseases in s...	Harder to enforce per-disease referen...	Add relation table (e.g., DiseaseSour...
Search requirements may need DB-level...	Text search may be slower at scale	Add PostgreSQL tsvector + GIN indexes...

References

- * backend/prisma/schema.prisma
- * backend/prisma/migrations/
- * backend/prisma/seed.ts
- * docker-compose.yml